



comp10002/comp20005

algorithms and asymptotics

Support material for Chapter 12 of
Programming, Problem Solving, and Abstraction with C,
by Alistair Moffat

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$O(1)$

constant

yes, Yes, YES!!!

$O(1)$

constant

yes, Yes, YES!!!

$O(\log \log n)$

VERY slow

yes, YES!!

$O(1)$

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VERY slow

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YES!

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VERY slow

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$O(\log n)$

very slow

YES!

$O(n^{1/2})$

slow

sweet as

$O(1)$	constant	yes, Yes, YES!!!
$O(\log\log n)$	VERY slow	yes, YES!!
$O(\log n)$	very slow	YES!
$O(n^{1/2})$	slow	sweet as
$O(n)$	linear	excellent

$O(1)$	constant	yes, Yes, YES!!!
$O(\log \log n)$	VERY slow	yes, YES!!
$O(\log n)$	very slow	YES!
$O(n^{1/2})$	slow	sweet as
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$O(n \log n)$	super linear	very good

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$O(n)$	linear	excellent
$O(n \log n)$	super linear	very good
$O(n^2)$	quadratic	ok, fair enough

$O(1)$	constant	yes, Yes, YES!!!
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$O(\log n)$	very slow	YES!
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$O(n \log n)$	super linear	very good
$O(n^2)$	quadratic	ok, fair enough
$O(n^3)$	cubic	tolerable

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$O(\log n)$	very slow	YES!
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$O(n^3)$	cubic	tolerable
$O(n^4)$	quartic	annoying

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$O(\log n)$	very slow	YES!
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$O(n)$	linear	excellent
$O(n \log n)$	super linear	very good
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$O(2^n)$	exponential	intractable

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$O(n)$	linear	excellent
$O(n \log n)$	super linear	very good
$O(n^2)$	quadratic	ok, fair enough
$O(n^3)$	cubic	tolerable
$O(n^4)$	quartic	annoying
$O(2^n)$	exponential	intractable
$O(2^{2^n})$	double exp	*&%#!@#^

if we have 10^{10} loop
iterations to spend...

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...something like 10 – 30
seconds of computation...

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iterations to spend...

...something like 10 – 30
seconds of computation...

...what value of n
can be reached?

$O(1)$	constant	yes, Yes, YES!!!
$O(\log \log n)$	VERY slow	yes, YES!!
$O(\log n)$	very slow	YES!
$O(n^{1/2})$	slow	sweet as
$O(n)$	linear	excellent
$O(n \log n)$	super linear	very good
$O(n^2)$	quadratic	ok, fair enough
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$O(2^{2^n})$	double exp	*&%#!@#^

$O(1)$	constant	yes, Yes, YES!!!	
$O(\log\log n)$	VERY slow	yes, YES!!	
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$O(n)$	linear	excellent	
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$O(n^3)$	cubic	tolerable	
$O(n^4)$	quartic	annoying	
$O(2^n)$	exponential	intractable	
$O(2^{2^n})$	double exp	*&%#!@#^	useless, 5

$O(1)$	constant	yes, Yes, YES!!!	
$O(\log \log n)$	VERY slow	yes, YES!!	
$O(\log n)$	very slow	YES!	
$O(n^{1/2})$	slow	sweet as	
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$O(n^3)$	cubic	tolerable	
$O(n^4)$	quartic	annoying	
$O(2^n)$	exponential	intractable	tiny, 33
$O(2^{2^n})$	double exp	*&%#!@#^	useless, 5

$O(1)$	constant	yes, Yes, YES!!!	
$O(\log \log n)$	VERY slow	yes, YES!!	
$O(\log n)$	very slow	YES!	
$O(n^{1/2})$	slow	sweet as	
$O(n)$	linear	excellent	
$O(n \log n)$	super linear	very good	
$O(n^2)$	quadratic	ok, fair enough	
$O(n^3)$	cubic	tolerable	
$O(n^4)$	quartic	annoying	small, 315
$O(2^n)$	exponential	intractable	tiny, 33
$O(2^{2^n})$	double exp	*&%#!@#^	useless, 5

$O(1)$	constant	yes, Yes, YES!!!	
$O(\log \log n)$	VERY slow	yes, YES!!	
$O(\log n)$	very slow	YES!	
$O(n^{1/2})$	slow	sweet as	
$O(n)$	linear	excellent	
$O(n \log n)$	super linear	very good	
$O(n^2)$	quadratic	ok, fair enough	
$O(n^3)$	cubic	tolerable	modest, 2150
$O(n^4)$	quartic	annoying	small, 315
$O(2^n)$	exponential	intractable	tiny, 33
$O(2^{2^n})$	double exp	*&%#!@#^	useless, 5

$O(1)$	constant	yes, Yes, YES!!!	
$O(\log \log n)$	VERY slow	yes, YES!!	
$O(\log n)$	very slow	YES!	
$O(n^{1/2})$	slow	sweet as	
$O(n)$	linear	excellent	
$O(n \log n)$	super linear	very good	
$O(n^2)$	quadratic	ok, fair enough	biggish, 10^5
$O(n^3)$	cubic	tolerable	modest, 2150
$O(n^4)$	quartic	annoying	small, 315
$O(2^n)$	exponential	intractable	tiny, 33
$O(2^{2^n})$	double exp	*&%#!@#^	useless, 5

$O(1)$	constant	yes, Yes, YES!!!	
$O(\log \log n)$	VERY slow	yes, YES!!	
$O(\log n)$	very slow	YES!	
$O(n^{1/2})$	slow	sweet as	
$O(n)$	linear	excellent	
$O(n \log n)$	super linear	very good	large, $10^{8.5}$
$O(n^2)$	quadratic	ok, fair enough	biggish, 10^5
$O(n^3)$	cubic	tolerable	modest, 2150
$O(n^4)$	quartic	annoying	small, 315
$O(2^n)$	exponential	intractable	tiny, 33
$O(2^{2^n})$	double exp	*&%#!@#^	useless, 5

$O(1)$	constant	yes, Yes, YES!!!	
$O(\log \log n)$	VERY slow	yes, YES!!	
$O(\log n)$	very slow	YES!	
$O(n^{1/2})$	slow	sweet as	
$O(n)$	linear	excellent	huge, 10^{10}
$O(n \log n)$	super linear	very good	large, $10^{8.5}$
$O(n^2)$	quadratic	ok, fair enough	biggish, 10^5
$O(n^3)$	cubic	tolerable	modest, 2150
$O(n^4)$	quartic	annoying	small, 315
$O(2^n)$	exponential	intractable	tiny, 33
$O(2^{2^n})$	double exp	*&%#!@#^	useless, 5

$O(1)$	constant	yes, Yes, YES!!!	
$O(\log\log n)$	VERY slow	yes, YES!!	
$O(\log n)$	very slow	YES!	
$O(n^{1/2})$	slow	sweet as	enormous, 10^{20}
$O(n)$	linear	excellent	huge, 10^{10}
$O(n \log n)$	super linear	very good	large, $10^{8.5}$
$O(n^2)$	quadratic	ok, fair enough	biggish, 10^5
$O(n^3)$	cubic	tolerable	modest, 2150
$O(n^4)$	quartic	annoying	small, 315
$O(2^n)$	exponential	intractable	tiny, 33
$O(2^{2^n})$	double exp	*&%#!@#^	useless, 5

$O(1)$	constant	yes, Yes, YES!!!	
$O(\log \log n)$	VERY slow	yes, YES!!	
$O(\log n)$	very slow	YES!	astronomical
$O(n^{1/2})$	slow	sweet as	enormous, 10^{20}
$O(n)$	linear	excellent	huge, 10^{10}
$O(n \log n)$	super linear	very good	large, $10^{8.5}$
$O(n^2)$	quadratic	ok, fair enough	biggish, 10^5
$O(n^3)$	cubic	tolerable	modest, 2150
$O(n^4)$	quartic	annoying	small, 315
$O(2^n)$	exponential	intractable	tiny, 33
$O(2^{2^n})$	double exp	*&%#!@#^	useless, 5

$O(1)$	constant	yes, Yes, YES!!!	
$O(\log\log n)$	VERY slow	yes, YES!!	to infinity
$O(\log n)$	very slow	YES!	astronomical
$O(n^{1/2})$	slow	sweet as	enormous, 10^{20}
$O(n)$	linear	excellent	huge, 10^{10}
$O(n \log n)$	super linear	very good	large, $10^{8.5}$
$O(n^2)$	quadratic	ok, fair enough	biggish, 10^5
$O(n^3)$	cubic	tolerable	modest, 2150
$O(n^4)$	quartic	annoying	small, 315
$O(2^n)$	exponential	intractable	tiny, 33
$O(2^{2^n})$	double exp	*&%#!@#^	useless, 5

$O(1)$	constant	yes, Yes, YES!!!	and beyond!
$O(\log\log n)$	VERY slow	yes, YES!!	to infinity
$O(\log n)$	very slow	YES!	astronomical
$O(n^{1/2})$	slow	sweet as	enormous, 10^{20}
$O(n)$	linear	excellent	huge, 10^{10}
$O(n \log n)$	super linear	very good	large, $10^{8.5}$
$O(n^2)$	quadratic	ok, fair enough	biggish, 10^5
$O(n^3)$	cubic	tolerable	modest, 2150
$O(n^4)$	quartic	annoying	small, 315
$O(2^n)$	exponential	intractable	tiny, 33
$O(2^{2^n})$	double exp	*&%#!@#^	useless, 5

are there more functions
in this spectrum?

are there more functions
in this spectrum?

sure, as many as you want!

$O(1)$

$O(\log \log n)$

$O(\log n)$

$O(n^{1/2})$

$O(n)$

$O(n \log n)$

$O(n^2)$

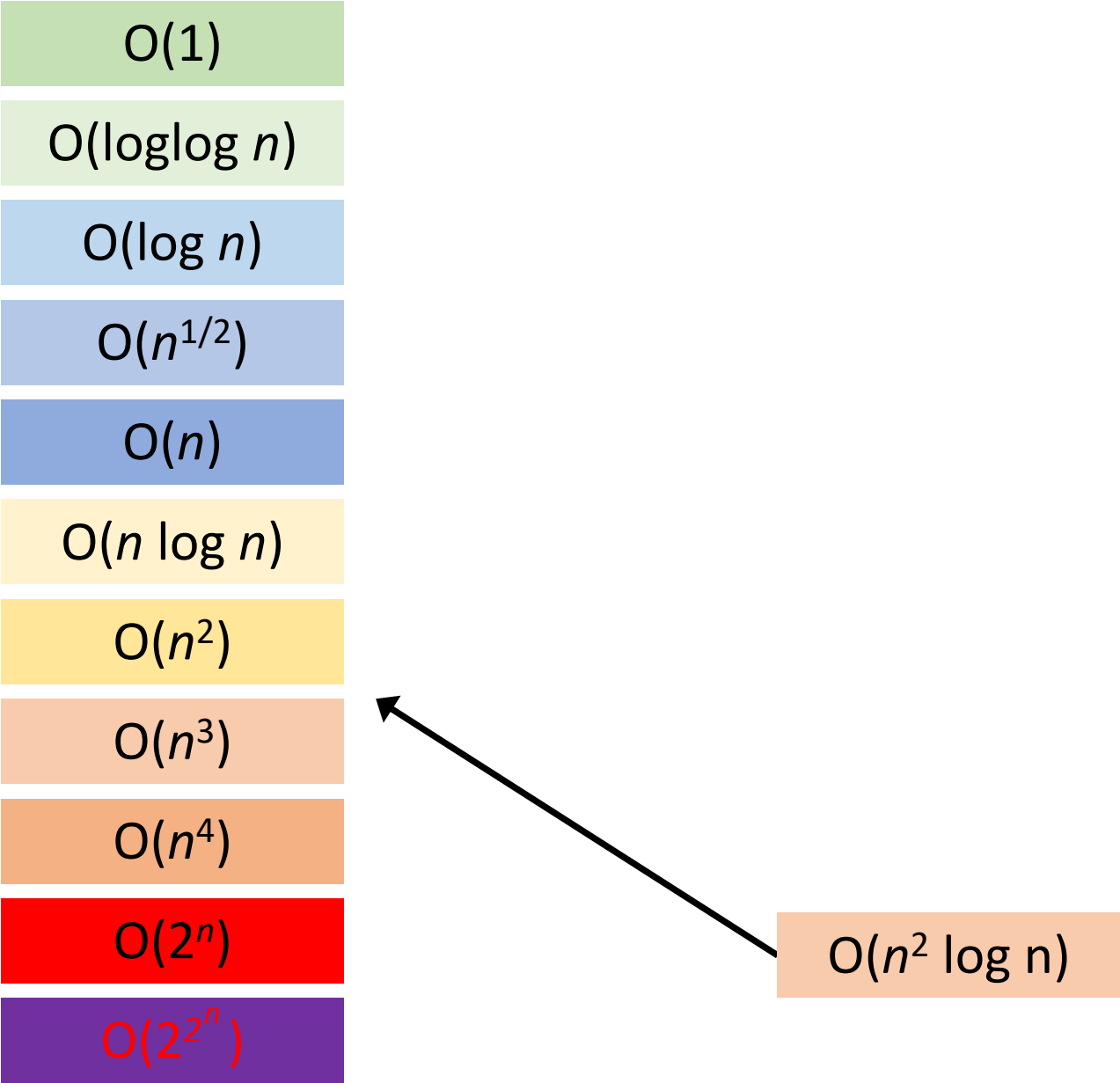
$O(n^3)$

$O(n^4)$

$O(2^n)$

$O(2^{2^n})$

$O(n^2 \log n)$



$O(1)$

$O(\log \log n)$

$O(\log n)$

$O(n^{1/2})$

$O(n)$

$O(n \log n)$

$O(n^2)$

$O(n^3)$

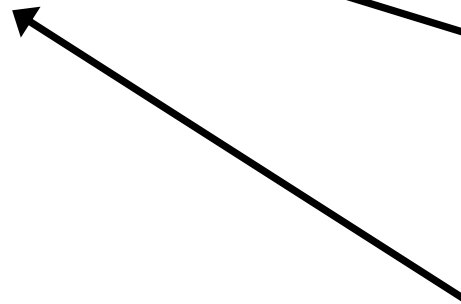
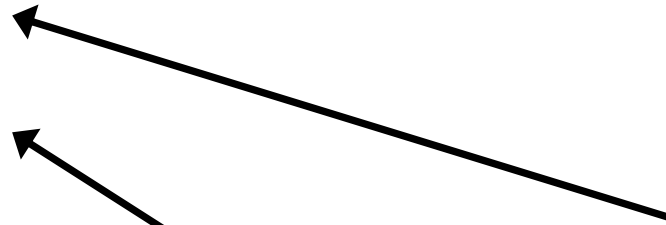
$O(n^4)$

$O(2^n)$

$O(2^{2^n})$

$O(n \log^2 n)$

$O(n^2 \log n)$



$O(1)$

$O(\log \log n)$

$O(\log n)$

$O(n^{1/2})$

$O(n)$

$O(n \log n)$

$O(n^2)$

$O(n^3)$

$O(n^4)$

$O(2^n)$

$O(2^{2^n})$

